PHASE 0 TECHNICAL NOTE · LATTICEMEM RESEARCH PROJECT

The Golay Cocode as the Dual Address Space for Turyn Glue

A syndrome-table decoder for the Leech lattice. The Turyn glue group — the 4096 cosets of the three √2–scaled E8 blocks — is elementary abelian and is resolved exactly by the Golay cocode syndrome σ together with a single mod-4 sum bit. The resulting decoder uses a 2^{12}-entry dual table, is provably exact within radius 1, and is certified against two independent ML oracles.

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Companion implementation: **leech_cocode** (Python, exact arithmetic)
All structural claims verified by exhaustive computation

Glue group G of order 4096 ≅ $(Z/2)^{12}$; integer part = ker σ ∩ ker Σ

Syndrome space 13 bits: (s, t) ∈ C* × Z/4 with t ≡ f(s) mod 2 — 12 of them from the cocode

Exactness proved within radius 1; 99.9–100% ML agreement inside the BDD ball

Phase 0 + Phase 1 + Phase 2 deliverable

November 3, 2026

Phase 0 Technical Note

The Golay Cocode as the Dual Address Space for Turyn Glue:

A Syndrome-Table Decoder for the Leech Lattice

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Abstract

The Turyn construction realizes the Leech lattice Λ_{24} as an overlattice of $\sqrt{2}E_8^3$, glued by a group of 4096 cosets. We show that the natural syndrome space for this glue is carried by the *Golay cocode* $\mathcal{C}^* = \mathbb{F}_2^{24}/\mathcal{G}_{24}$. Concretely, on the ambient class group $\mathcal{A} = (\mathbb{Z}^8/2E_8)^3$ there are two computable homomorphisms: the cocode syndrome $\sigma : \mathcal{A} \to \mathcal{C}^*$ and the mod-4 sum syndrome $\Sigma : \mathcal{A} \to \mathbb{Z}/4$, and the integer part of the glue group is exactly $G_{\text{int}} = \ker\sigma \cap \ker\Sigma$. The full glue group has order $|\mathcal{C}^*| = 4096$, is elementary abelian $(\mathbb{Z}/2)^{12}$—the same group type as the cocode—and the quotient $\mathcal{A}/G$ is the 13-bit space $\{(s,t) \in \mathcal{C}^* \times \mathbb{Z}/4 : \ t \equiv f(s) \bmod 2\}$, where f is the weight parity of a coset representative. The cocode thus supplies 12 of the 13 syndrome bits. This yields a decoder that rounds, computes one cocode syndrome and one sum, then corrects through a precomputed 2^{12}-entry coset table with sign matching and a single ± 2 adjustment. The decoder is provably exact within radius 1 (in $\sqrt{2}\Lambda_{24}$ scale), never returns sub-optimal points, and attains 99.9–100% agreement with exact maximum-likelihood decoding inside the bounded-distance ball; a candidate-weight knob W trades empirical ML agreement against speed (96.3%/98.7%/99.1% at $W = 4/5/6$ under $\sigma = 0.2$ noise). All structural claims are verified by exact integer computation, and all decoding claims by cross-validation against two independent exact ML oracles that agree to 10^{-15}.

1 Introduction and Summary of Results

The Leech lattice Λ_{24} is the unique even unimodular lattice in $\mathbb{R}^{24}$ with minimum norm 4 [1]. Its decoding is a classical problem at the intersection of coding theory and lattice algorithms: Be'ery and Vardy developed the standard maximum-likelihood (ML) decoder [2], and a neural decoder built on the Turyn construction was studied by Corlay, Joseph and Boutros [3, 4]. The Turyn construction is one of the 23 constructions catalogued by Sloane [5]: it realizes Λ_{24} as an overlattice of three copies of $\sqrt{2}E_8$, with the three blocks coupled by a glue condition. The present note is the Phase 0 deliverable of a research programme whose guiding idea is that the *Golay cocode* $\mathcal{C}^* = \mathbb{F}_2^{24}/\mathcal{G}_{24}$—the syndrome space of the extended binary Golay code $\mathcal{G}_{24}$—can serve as the *dual address space* for the Turyn glue, so that glue correction reduces to a table lookup indexed by a cocode syndrome.

That idea is correct, with one refinement that we regard as the main contribution of this note: the cocode carries 12 of the 13 bits needed to resolve the glue, and the missing bit is the mod-4 parity of the coordinate sum. Everything is stated on an explicitly computed model, and every claim below was verified by exact integer arithmetic; nothing rests on unverified recall of the literature. Our results are:

1. **The Turyn filtration, made explicit** (Theorem 1). With a suitable "trio" labeling of the Golay coordinates by three octads, the sublattice $2E_8^3 \subset \mathbb{Z}^{24}$ (of points $\sqrt{2}E_8^3$ in the $\sqrt{2}$-scaled model of Λ_{24}) has index 4096 in $\sqrt{2}\,\Lambda_{24}$. The glue group $G = \sqrt{2}\Lambda_{24}/2E_8^3$ is elementary abelian of order $4096 = 2^{12}$.

2. **The syndrome theorem** (Theorem 5). On the ambient class group $\mathcal{A} = (\mathbb{Z}^8/2E_8)^3$ of order 2^{24} there are two natural homomorphisms: the cocode syndrome σ and the sum syndrome Σ. The integer part of the glue group is *exactly* $\ker\sigma \cap \ker\Sigma$, and the full quotient $\mathcal{A}/G$ is the space of pairs $(s,t) \in \mathcal{C}^* \times \mathbb{Z}/4$ with $t \equiv f(s) \pmod 2$, of order 8192.
3. **The Cocode–Turyn dual decoder** (Section 5). A decoding algorithm organized around the cocode address: round, compute s and t, look up the coset-leader table (2^{12} entries, average 3.16 candidates of weight ≤ 4), match signs, and apply one ± 2 sum fix, in two shell passes. It is provably exact within radius 1 and empirically near-ML elsewhere; it never returns a point closer than the ML distance.
4. **Verification** (Section 6). Two independent exact ML oracles (family enumeration and glue-coset enumeration with per-block E_8 decoding) agree to 10^{-15}; the decoder matches them on 10^4 random inputs with the statistics reported in Table 1.

A noteworthy structural finding is negative: the glue group does *not* split as a product of per-block groups. Its per-block projections have order 128, and the "block-diagonal" intuition behind a naive per-block decoder fails; the naive Turyn coset enumeration costs $3 \cdot 128$ exact E_8 decodes per point, whereas the cocode syndrome diagonalizes the coupling at the cost of one syndrome computation (Section 5.2). This is precisely the sense in which the cocode "decouples the syndrome computation from the lattice construction."

2 Notation and Constructions

2.1 The Golay code and the trio labeling

We construct the extended binary Golay code $\mathcal{G}_{24}$ from scratch as the extended quadratic-residue code of length 23: over $GF(2^{11})$, with α a primitive 23rd root of unity, the generator polynomial $g(x) = \prod_{i\in Q}(x-\alpha^i)$, Q the nonzero quadratic residues mod 23, has coefficients in $\mathbb{F}_2$ (the residue set is a single 2-cyclotomic coset), and the cyclic code it spans has dimension 12. Appending an overall parity bit yields a $[24, 12]$ code whose weight enumerator we verify to be

$$1,\ 759,\ 2576,\ 759,\ 1 \quad \text{at weights } 0, 8, 12, 16, 24,$$

and whose generators are pairwise orthogonal, i.e. the code is self-dual. This is the Golay code. Every word has weight divisible by 4; the 759 words of weight 8 are the *octads*.

We relabel coordinates by a *trio*: three pairwise disjoint octads b_1, b_2, b_3 partitioning the 24 coordinates. After relabeling, the block indicators

$$\mathbf{1}_{b_1} = (1^8, 0^{16}), \qquad \mathbf{1}_{b_2} = (0^8, 1^8, 0^8), \qquad \mathbf{1}_{b_3} = (0^{16}, 1^8)$$

and their span (the 8-element "repetition" subcode) lie in $\mathcal{G}_{24}$. All subsequent constructions use this labeling.

2.2 The Leech lattice in integer coordinates

Work in the $\sqrt{2}$-scaled model $\widetilde{\Lambda_{24}} = \sqrt{2}\,\Lambda_{24}$, so that lattice points have integer or half-odd coordinates. Define

$$M_0 = \{x \in \mathbb{Z}^{24} :\ x \bmod 2 \in \mathcal{G}_{24},\ \ \textstyle\sum_i x_i \equiv 0 \bmod 4\}, \qquad \tau = \tfrac{1}{2}(-3, 1^{23}),$$

and $\widetilde{\Lambda_{24}} = M_0 \cup (\tau + M_0)$. One checks that $2\tau = (-3, 1^{23})$ lies in M_0, so $\widetilde{\Lambda_{24}}$ is an additive group. We verified computationally, exactly:

- *Evenness.* M_0 and $\tau + M_0$ have all squared norms divisible by 4; the mixed inner products are integral (for $z \in M_0$, $z \cdot 2\tau \equiv \sum z_i \equiv 0 \bmod 2$).

- *Minimum norm 4 and kissing number.* The norm-4 vectors of M_0 are exactly the shapes $(\pm 2^2, 0^{22})$ $(1104 = 4\binom{24}{2})$ and $(\pm 1^8, 0^{16})$ on octads with sign patterns whose sum is 0 mod 4 $(759 \cdot 128 = 97152)$; the norm-4 vectors of $\tau + M_0$ are exactly the shapes $\frac{1}{2}(\mp 3, \pm 1^{23})$, one for each pair (codeword, position) $(4096 \cdot 24 = 98304)$. The total is $1104 + 97152 + 98304 = 196560$.
- *Determinant.* The index $[\widetilde{\Lambda_{24}} : 2E_8^3]$ equals 4096 (breadth-first closure over class representatives, Section 3); since $\det(2E_8^3) = 2^{48}$, we get $\det \widetilde{\Lambda_{24}} = 2^{48}/4096^2 = 2^{24}$ and hence $\det \Lambda_{24} = 2^{24}/(\sqrt{2})^{48} = 1$.

Even, unimodular, minimum 4 in dimension 24 characterizes Λ_{24} uniquely [1], so the model is the Leech lattice.

2.3 The E_8 blocks

Inside one 8-coordinate block we use

$$2E_8 \;=\; \{\, v \in \mathbb{Z}^8 :\; v \bmod 2 \in \{0^8, 1^8\},\;\; \textstyle\sum_i v_i \equiv 0 \bmod 4 \,\} \;=\; 2D_8 \,\cup\, (2D_8 + \mathbf{1}),$$

a lattice of determinant 2^{16}, so $[\mathbb{Z}^8 : 2E_8] = 2^8 = 256$. The generator containment $2E_8^3 \subset M_0$ is immediate from the definitions (block indicators are codewords; block sums are 0 mod 4) and was verified on generators. This is the $\sqrt{2}E_8^3 \subset \Lambda_{24}$ filtration of the Turyn construction: in the original scale the three blocks are copies of $\sqrt{2}E_8$ and Λ_{24} is the union of 4096 of their cosets.

3 The Glue Group and the Ambient Class Group

3.1 The ambient class group

Let $\mathcal{A} = \mathbb{Z}^{24}/2E_8^3 \cong (\mathbb{Z}^8/2E_8)^3$, of order 2^{24}. A complete invariant of the class of $v \in \mathbb{Z}^8$ modulo $2E_8$ is the pair

$$\big(\overline{v \bmod 2},\;\; \textstyle\sum v_i \bmod 4\,\big),$$

where the bar denotes the parity vector modulo the all-flip $v \mapsto v + \mathbf{1}$ (adding $\mathbf{1} \in 2E_8$ changes no sums, since $\sum \mathbf{1} = 8 \equiv 0 \bmod 4$). Indeed two vectors with equal invariants differ by an element of $2D_8 \cup (2D_8 + \mathbf{1}) = 2E_8$. Each block thus contributes $2^7 \cdot 2 = 2^8$ classes, and the class of e_1 has order 4 (since $2e_1 \notin 2E_8$ but $4e_1 \in 2E_8$), so $\mathcal{A}$ is not elementary abelian. The *full* ambient group also records the shell: every point of $\tau + \mathbb{Z}^{24}$ defines a class, and since $2\tau \in 2E_8^3$,

$$\mathcal{A}_{\text{full}} \;=\; \mathcal{A} \;\times\; \langle [\tau] \rangle, \qquad |\mathcal{A}_{\text{full}}| = 2^{25},$$

with $[\tau]$ of order 2.

3.2 The glue group

The *Turyn glue group* is $G = \widetilde{\Lambda_{24}}/2E_8^3 \subset \mathcal{A}_{\text{full}}$.

Theorem 1 (Turyn filtration; verified)**.** $2E_8^3 \subset \widetilde{\Lambda_{24}}$ *with index* 4096*. The glue group G splits as $G = G_{\text{int}} \sqcup ([\tau] + G_{\text{int}})$ with $|G_{\text{int}}| = 2048$, and every element of G has order at most 2; hence*

$$G \;\cong\; (\mathbb{Z}/2)^{12},$$

the same abelian group type as the cocode $\mathcal{C}^* = \mathbb{F}_2^{24}/\mathcal{G}_{24}$.

The order was computed by breadth-first closure of the class-group arithmetic over the generators of M_0 together with τ; the element orders were computed exhaustively. The per-block projections were also computed:

Proposition 2 (verified)**.** *For each block i, $|\pi_i(G)| = 128$, and the pairwise projections have order* 2048*. Consequently G is not a product of per-block groups: the glue constraint couples the blocks strictly.*

This is the structural reason a literal per-block decoder ("decode each E_8 block, then fix the glue") is expensive: the per-block alphabets have 128 classes, and assembling them under the glue constraint naively requires $3 \cdot 128$ E_8-coset decodes per point. The remedy is the syndrome formulation of Section 4, which replaces the block-coupled search by one homomorphism evaluation.

4 The Cocode–Turyn Duality Theorem

Definition 3 (Cocode syndrome and sum syndrome)**.** For a class $[x] \in \mathcal{A}$ define

$$\sigma([x]) \;=\; (x \bmod 2) + \mathcal{G}_{24} \;\in\; \mathcal{C}^*, \qquad \Sigma([x]) \;=\; \textstyle\sum_i x_i \bmod 4 \;\in\; \mathbb{Z}/4.$$

Lemma 4 (well-definedness; verified)**.** *Both maps are well defined homomorphisms. For σ this uses the fact that the ambiguity of the parity representative—the block-flip subgroup $\{0,1\}^8$ per block—is contained in the trio subcode of $\mathcal{G}_{24}$, so the cocode coset is unaffected. For Σ it uses $\sum(\mathbb{Z}^8 \cap 2E_8) \equiv 0 \bmod 4$. On the τ-shell we set $\sigma([\tau + z]) = \sigma([z])$ and $\Sigma([\tau + z]) = \Sigma([z])$; since $2\tau \in 2E_8^3$ with $\Sigma([2\tau]) = 0$, this extends the two maps to $\mathcal{A}_{\text{full}}$.*

The homomorphism property was verified on random samples; well-definedness is the elementary argument just given, and both were exercised inside all downstream exhaustive checks.

Theorem 5 (syndrome theorem; verified)**.**

1. $G_{\text{int}} = \ker\sigma \cap \ker\Sigma$ *inside $\mathcal{A}$. Both sides have order* 2048*, and each was enumerated independently: the left side by closure over M_0-generators, the right side by direct enumeration of canonical parities coming from codewords with admissible sums. The two sets coincide.*
2. *Let $f : \mathcal{C}^* \to \mathbb{Z}/2$ be the weight parity of any representative of the coset (well defined because all codewords are even). Then*

$$\mathcal{A}_{\text{full}}/G \;\cong\; \{\,(s,t) \in \mathcal{C}^* \times \mathbb{Z}/4 :\; t \equiv f(s) \bmod 2\,\},$$

   a space of order $2^{13} = 8192 = 2\,|\mathcal{C}^*|$. The isomorphism is induced by $[x] \mapsto (\sigma([x]), \Sigma([x]))$, whose kernel is exactly G.
3. $\ker\sigma = G \times \langle [v] \rangle$ *for any class $[v]$ with $\sigma([v]) = 0$ and $\Sigma([v]) = 2$, e.g. the class of $c + 2e_1$ for a codeword c; in particular G has index* 2 *in the cocode kernel.*

Proof sketch. The relation $t \equiv f(s)$ reflects that $\sum x_i \equiv \mathrm{wt}(x \bmod 2) \bmod 2$, and the weight parity is a function of the cocode coset. Counting then forces everything: $|\mathcal{A}_{\text{full}}| = 2^{25}$, the constrained pair space has 2^{13} elements, and the kernel, computed as $\ker\sigma \cap \ker\Sigma$ on the integer side together with its τ-translate, has $2^{25}/2^{13} = 2^{12}$ elements, which the enumeration identifies with G. All steps were carried out exactly; the two sides of item 1 are independent computations that agree, which is the strongest form of verification available for this statement. □

Remark 6 (the precise sense of the duality). The original research plan conjectured an injective map $\phi\colon G \to \mathcal{C}^*$. The truth is subtler and, we believe, better: there is no natural injection of the glue group into the cocode (the cocode syndromes of glue representatives all vanish), but the cocode *addresses the correction problem*: the coset of possible corrections carrying a given glue syndrome is a translate of G indexed by the pair (s,t), of which the cocode supplies s. The 2^{12}-entry table of the plan exists—it is the Golay coset-leader table—and the extra bit t is handled by a single ± 2 adjustment rule rather than by table enlargement. Also, the frequently cited count "4096 cosets of $\sqrt{2}E_8^3$" is confirmed exactly, while the "16 per block" heuristic is not: the cube root 16 of $|G|$ does not correspond to any per-block splitting (Proposition 2); the per-block projections have order 128.

5 The Cocode–Turyn Dual Decoder

5.1 Algorithm

Input $y \in \mathbb{R}^{24}$ (coordinates in the $\widetilde{\Lambda_{24}}$ scale). The decoder runs two *shell passes* (integer shell, τ-shell) and returns the better point.

1. **Round.** $z = \mathrm{round}(y)$, $\rho = y - z \in [-\frac{1}{2}, \frac{1}{2}]^{24}$.
2. **Syndromes.** $p = z \bmod 2$; the cocode address $s = \sigma([z]) \in \mathcal{C}^*$ (a 12-bit syndrome of p with respect to a self-dual parity-check basis); the sum bit pair $t = \sum z_i \bmod 4$.
3. **Table lookup.** Retrieve cand[s]: all words $w \in \mathbb{F}_2^{24}$ of weight $\le W$ in the coset s ($W = 4$ by default; average 3.16 candidates, since the covering radius of $\mathcal{G}_{24}$ is 4).
4. **Candidate correction.** For each w: set $\delta_i = \mathrm{sign}(\rho_i)$ on $\mathrm{supp}(w)$ and $\delta_i = 0$ elsewhere; if $\sum(z + \delta) \not\equiv 0 \bmod 4$, add a single ± 2 at the cheapest off-support position (toward ρ). Candidate $x = z + \delta$; accumulate $|y - x|^2$.
5. **Shell repeat.** Repeat steps 1–4 with $y - \tau$; add τ to the result.
6. **Select.** Return the candidate of minimum distance.

The correctness of the per-candidate step is elementary: for a fixed target parity class $c \in \mathcal{G}_{24}$ and fixed shell, the family $\{x \equiv c \bmod 2,\ \sum x \equiv r \bmod 4\}$ has, near y, the componentwise-closest member plus (at most) one ± 2 adjustment, because each unit move of size 2 changes the sum by 2 mod 4 and strictly increases the distance.

Theorem 7 (exactness radius; proved)**.** *If the nearest point $\hat{x} \in \widetilde{\Lambda_{24}}$ to y satisfies $\|y - \hat{x}\| < 1$, the decoder returns a point at distance $\|y - \hat{x}\|$ (equal to the ML distance). In particular decoding is unique and exact inside the ball of radius 1 (which is radius $1/\sqrt{2}$ in the Λ_{24} scale).*

Proof. Write $\hat{x} = z + \delta$ with $z = \mathrm{round}(y)$. Every coordinate with δ_i odd contributes at least $(\frac{1}{2})^2$ to $\|y - \hat{x}\|^2$, hence at most three coordinates can be odd when the total is < 1, and no ± 2 adjustment can occur (a ± 2 move alone costs $(2 - \frac{1}{2})^2 > 1$). Thus the parity error $w = \delta \bmod 2$ has weight $\le 3 \le 4$, so $w \in$ cand[s] and the decoder's candidate set contains the family of $\hat{x}$; within that family the per-candidate rule is exact (argument above), so the decoder's output distance is at most $\|y - \hat{x}\|$. It is a lattice distance, so it is at least the ML distance, which equals $\|y - \hat{x}\|$ by assumption. The τ-shell pass is identical, applied to $y - \tau$ and $\hat{x} - \tau$. □

5.2 Complexity

Under an add/multiply/compare model, one decode with $W = 4$ costs: 24 comparisons (rounding), 24 integer additions (sum), 12×24 bit operations or 24 table-XORs (syndrome), 24 operations for the sorted fix slots, and about 6.3 candidate evaluations (two shells, average 3.16 candidates) at roughly 12 real operations each with incremental distance bookkeeping, plus 24 for the shell comparison: *circa* 250 *real operations* and two table lookups. The table has 2^{12} entries. For comparison, the naive Turyn coset enumeration performs $3 \cdot |\pi_i(G)| = 384$ exact E_8 decodes (measured at ≈170 ms per point in our Python implementation), and the classical ML decoder of Be'ery and Vardy [2] remains the reference for exact ML decoding at bounded cost. We emphasize that operation counts across papers are only comparable under a common operation model, and we state ours explicitly rather than adopting counts from secondary sources.

6 Computational Verification and Benchmarks

6.1 Two independent exact oracles

Oracle A enumerates all 8192 families $\{x \equiv c \bmod 2,\ \sum x \equiv r \bmod 4\}$ over codewords c and both shells, and returns the family-best point (componentwise closest plus one ± 2). Oracle B enumerates all 4096 glue cosets, lifting explicit coset representatives, and performs an exact E_8 nearest-point decode per block (D_8 and $D_8 + \frac{1}{2}\mathbf{1}$ candidates with a single parity flip). On all test sets the two oracles agree to 1.3×10^{-15} in squared distance. The decoder's output distance is never below the oracle distance (minimum difference -6.7×10^{-16}, i.e. floating-point noise), confirming that it always returns genuine lattice points.

6.2 Exactness tests

We decode 10^4 random inputs per noise level, with lattice points drawn uniformly from the generators and Gaussian noise of standard deviation σ in the Λ_{24} scale. "BDD ball" means ML distance $< \sqrt{2}$ in the $\widetilde{\Lambda_{24}}$ scale (unique-decoding region).

Table 1: Exactness of the Cocode–Turyn decoder ($W = 4$) against exact ML decoding, 10^4 random inputs per row.

noise σ	ML agreement	in-BDD fraction	in-BDD agreement
0.20	96.15%	59.2%	99.93%
0.35	56.10%	0.25%	100%
0.50	54.38%	0.22%	100%

Inside the provable radius-1 region the agreement is exactly 100% in every run, as Theorem 7 predicts. The residual in-BDD misses at $\sigma = 0.2$ (0.07% of in-BDD points) occur near the outer shell of the BDD ball, where maximum-weight (5–7) parity errors become geometrically possible although still sub-optimal in most draws; this is the boundary behavior one expects from a leader-based candidate set.

6.3 The candidate-weight knob

Extending the candidate table to words of weight $\leq W$ trades exactness against speed. Table 2 reports a 3000-point sweep.

Table 2: Exactness/speed tradeoff via the candidate weight bound W (3000 points per configuration; pure-Python timing per point).

W	ML agreement (σ=0.2)	ML agreement (σ=0.35)	time
4	96.3%	54.3%	53 μs
5	98.7%	79.2%	131 μs
6	99.1%	89.9%	448 μs

The remaining gap at large noise is concentrated in parity errors of weight ≥ 7, which dominate deep noise; closing it fully requires the weighted-Golay candidate machinery of the classical ML decoders [2], which is orthogonal to the glue-syndrome contribution of this note.

6.4 Reference timings

In our unoptimized Python implementation: decoder W=4: 52–61 μs per point; vectorized family oracle: 88 μs per point; naive glue-coset oracle (the literal per-block Turyn search): $\approx$ 170 ms

per point. These numbers illustrate structural, not absolute, costs; the syndrome formulation removes a factor of $\sim 10^3$ relative to the naive Turyn enumeration in the same implementation.

7 Scope, Limitations, and Next Steps

What is established. All structural statements (Theorems 1 and 5, Proposition 2) are verified by exact integer computation from first principles—the Golay code is constructed, not assumed—and the decoder claims are cross-validated against two independent exact oracles. The exactness radius (Theorem 7) is proved.

What is not yet established. (i) Beyond the BDD ball the decoder is not exactly ML; the empirical agreement rates of Table 2 quantify the gap but a tight analytic characterization of the failure set (parity errors of weight 5–7 within radius $\sqrt{2}$) is open. (ii) The operation-count comparison with the classical decoder [2] needs a common operation model and a C/hardware implementation; we deliberately avoid adopting literature counts we could not verify against primary sources. (iii) The *holy construction* of Λ_{24} glues by the octacode over $\mathbb{Z}_4$ rather than by the binary Golay cocode [5, 1]; whether an analogous dual-table syndrome theory exists there—with the octacode's $\mathbb{Z}_4$-linear structure replacing the cocode—is the natural Phase 3 boundary test. Our expectation is that the sum-syndrome bit t is replaced by a $\mathbb{Z}_4$-valued residue carried by the octacode itself, but this is conjecture at present. (iv) The *permuted Turyn* variant and the dependence of the syndrome table on the choice of trio (the automorphism group M_{24} acts transitively on trios, so all choices are equivalent up to relabeling) deserve an explicit treatment.

Roadmap alignment. This note completes Phase 0 (mathematical ground truth and algorithm specification) and Phase 1 (working implementation with an exactness test suite on 10^4 inputs), and delivers the Phase 2 complexity count and benchmark table. The remaining work for a submission is Phase 3 (boundary tests for the holy and permuted constructions) and the paper itself.

References

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